



ROHDE & SCHWARZ

EMF System

TV Test Receiver

EMFT

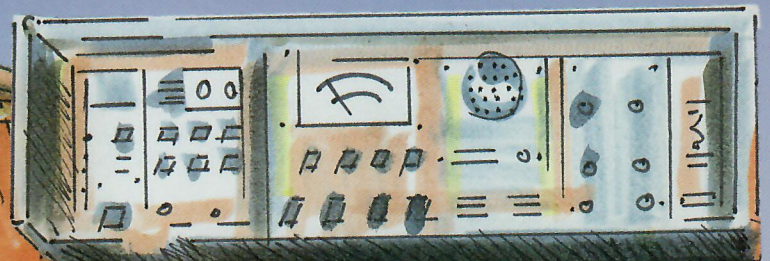
TV Test Demodulator

EMFD

TV Channel Receiver

EMFK

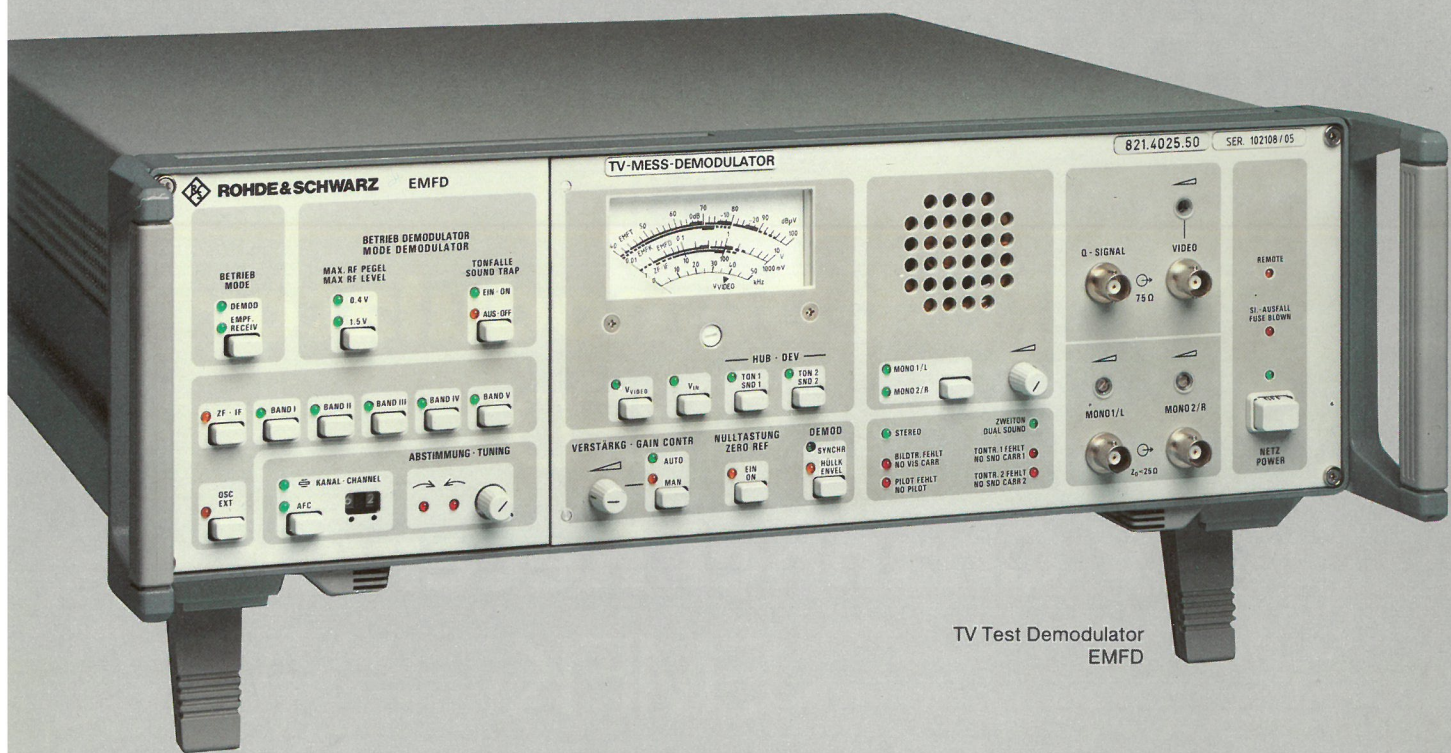
Standard B/G



Optional **IEC 625Bus** IEEE 488

- Real synchronous detection with quadrature signal output for measurement of incidental phase modulation of vision carrier
- Multitone capability in line with dual-carrier technique and NICAM
- RF input voltage indication
- Tunable in all TV bands including hyperband 330 to 470 MHz (EMFT)

Data sheet
821 401
E - 1



TV Test Demodulator EMFD

The new equipment family EMF is capable of solving all problems in the field of TV reception and demodulation. State-of-the art techniques such as synthesizer, double superheterodyning, SAW filter afford great operating convenience and outstanding transmission quality.

There are three units available for the different measurement tasks and transmission requirements:

- EMFT — continuously tunable test receiver and demodulator
- EMFD — TV test demodulator with AFC and crystal-controlled operation
- EMFK — selective, crystal-controlled channel receiver and demodulator

All units of the EMF family are provided with two isolated video outputs and two outputs for the Q signal. These outputs permit the measurement of the incidental phase modulation of the vision carrier which determines the intercarrier S/N ratio of the sound channel. In addition, the adjustment of linearity and phase equalizing circuits in transmitters is facilitated through the use of the Q signal.

The complex sound processing circuitry employs state-of-the-art ICs taking full account of present-day quality requirements. The deviation of the two sound demodulators can thus be guaranteed to be equal — which is the prerequisite for low stereo crosstalk.

Switchover from synchronous detection to envelope detection is possible. The input voltage and the frequency deviation of the sound carriers are indicated on the analog meter. The sound level is monitored by means of a built-in loudspeaker.

Special features

TV Test Receiver EMFT

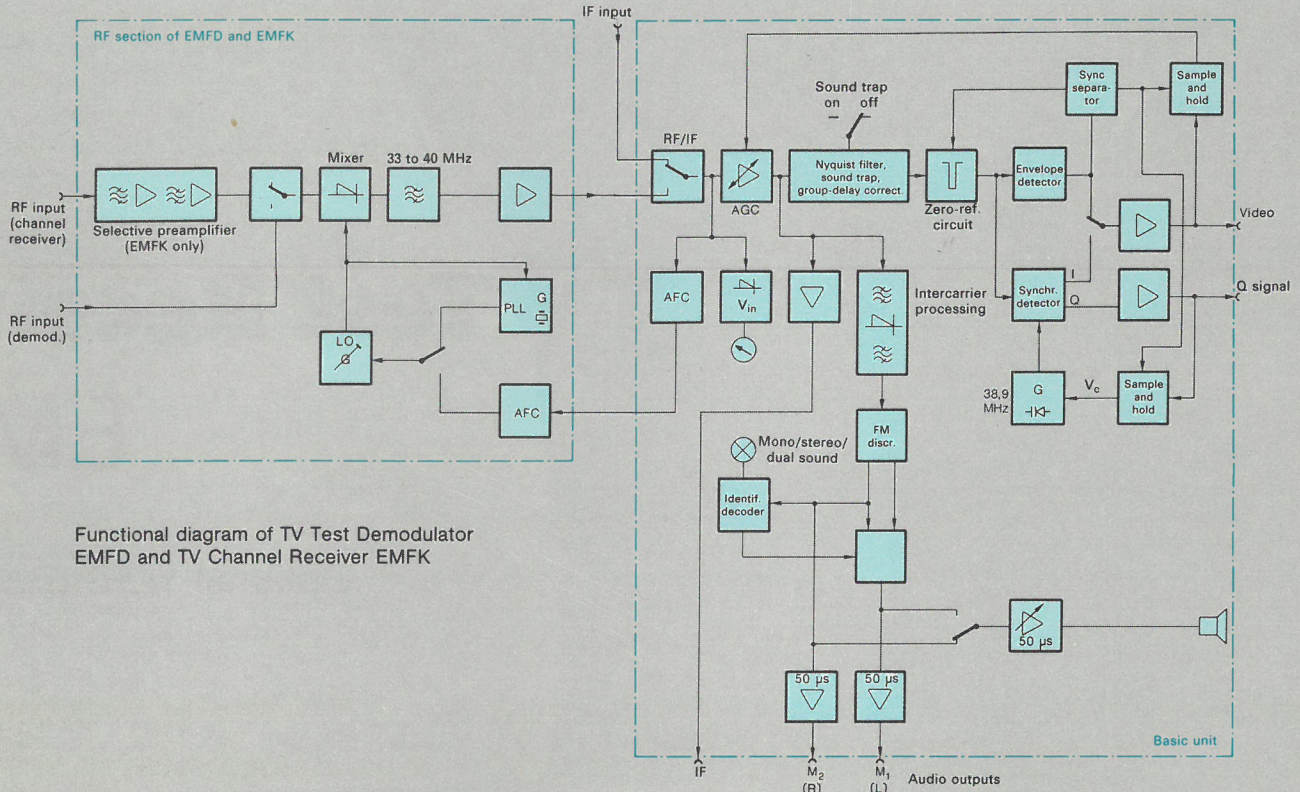
- Channel selection with digital entry of channel number either in manual or in remote mode
- Automatic control of frequency offset up to ± 100 kHz
- Continuously tunable in all TV bands, also in special channels up to 470 MHz
- SAW filter switch-selected on front panel for adjacent-channel suppression
- Optional IEC/IEEE-bus control (IEC 625-1/IEEE 488), thus channel selection also possible by entering the vision carrier frequency
- Reduction of IF gain allowing operation at high input levels (demodulator mode) and thus providing increased video S/N ratio

TV Test Demodulator EMFD

- Test demodulator for VHF, UHF and IF
- Crystal-controlled operation at a fixed frequency, switchover to continuously tuned AFC possible
- Sound trap can be switched off
- Selectable input sensitivity
- IF output suitable for connection of TV Dual-sound Demodulator FATF for deviation measurement

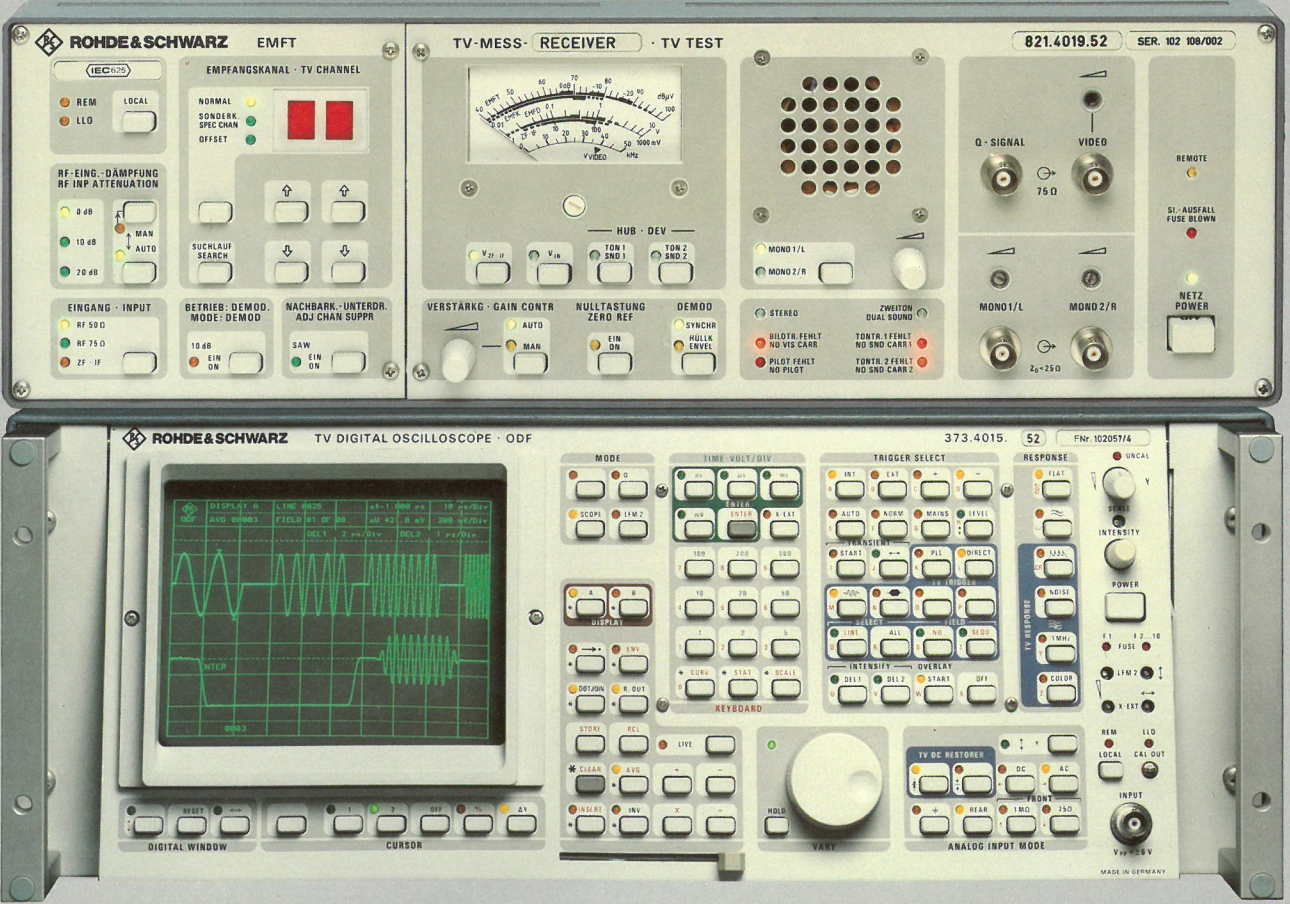
TV Channel Receiver EMFK

- Suitable as relay receiver for transmission and measurements
- Continuously tunable demodulator same as EMFD, however, with additional input via selective filter and preamplifier



Functional diagram of TV Test Demodulator EMFD and TV Channel Receiver EMFK

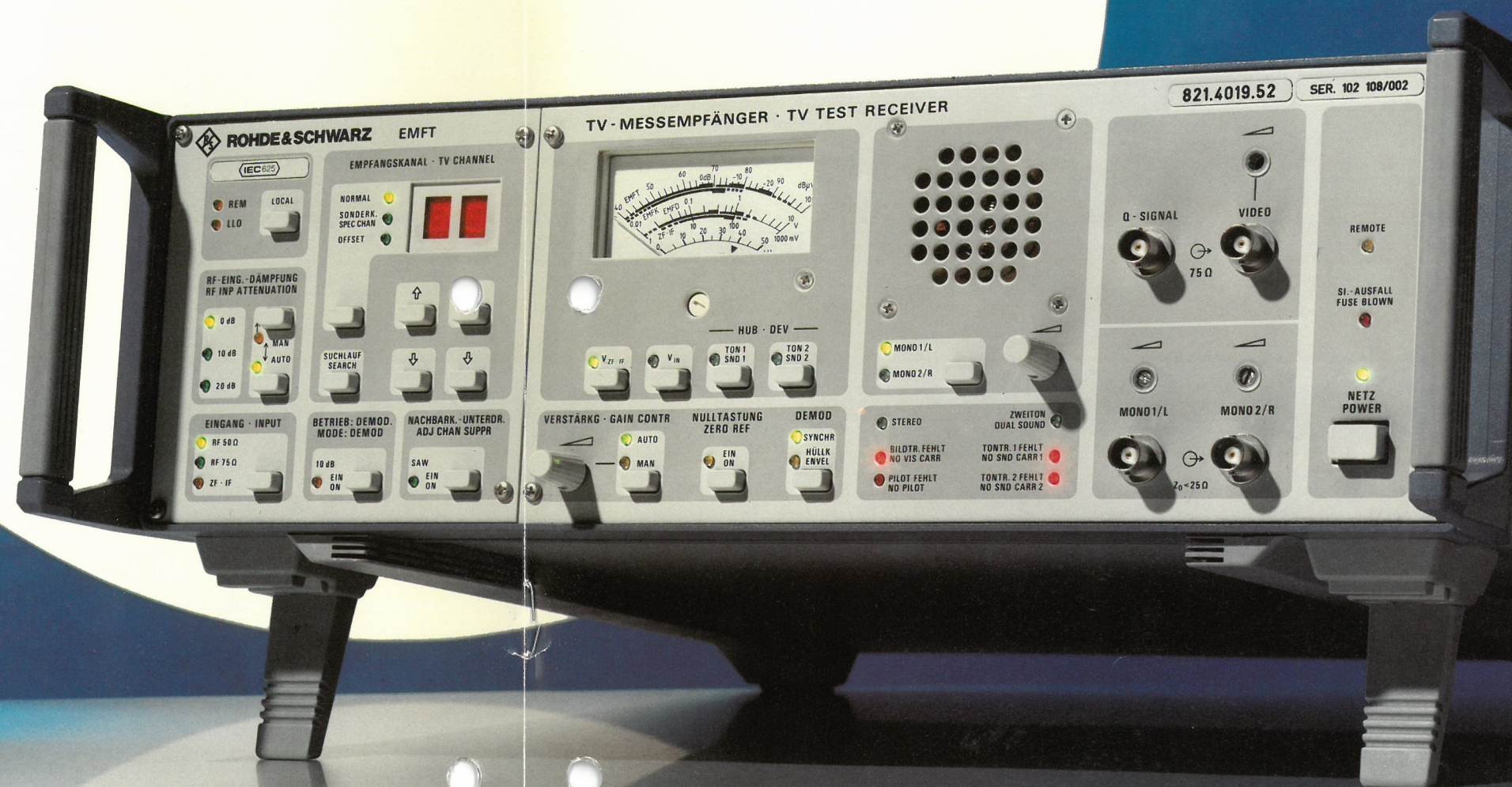
TV Test Receiver EMFT (top) plus TV Digital Oscilloscope ODF used for insertion signal measurement



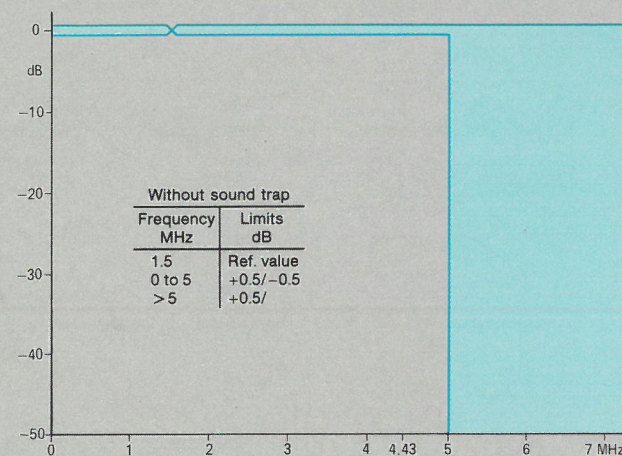
TV Test Receiver

EMFT

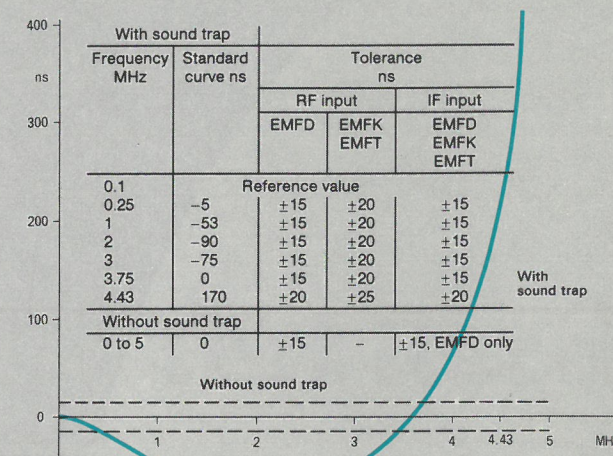
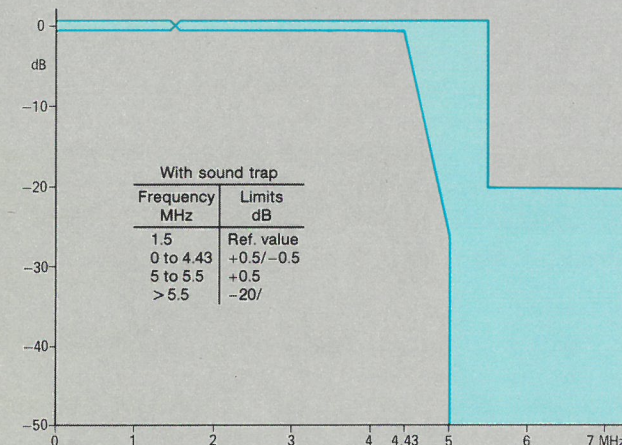
Optional **IEC 625 Bus** IEEE 488



TOLERANCE MASKS

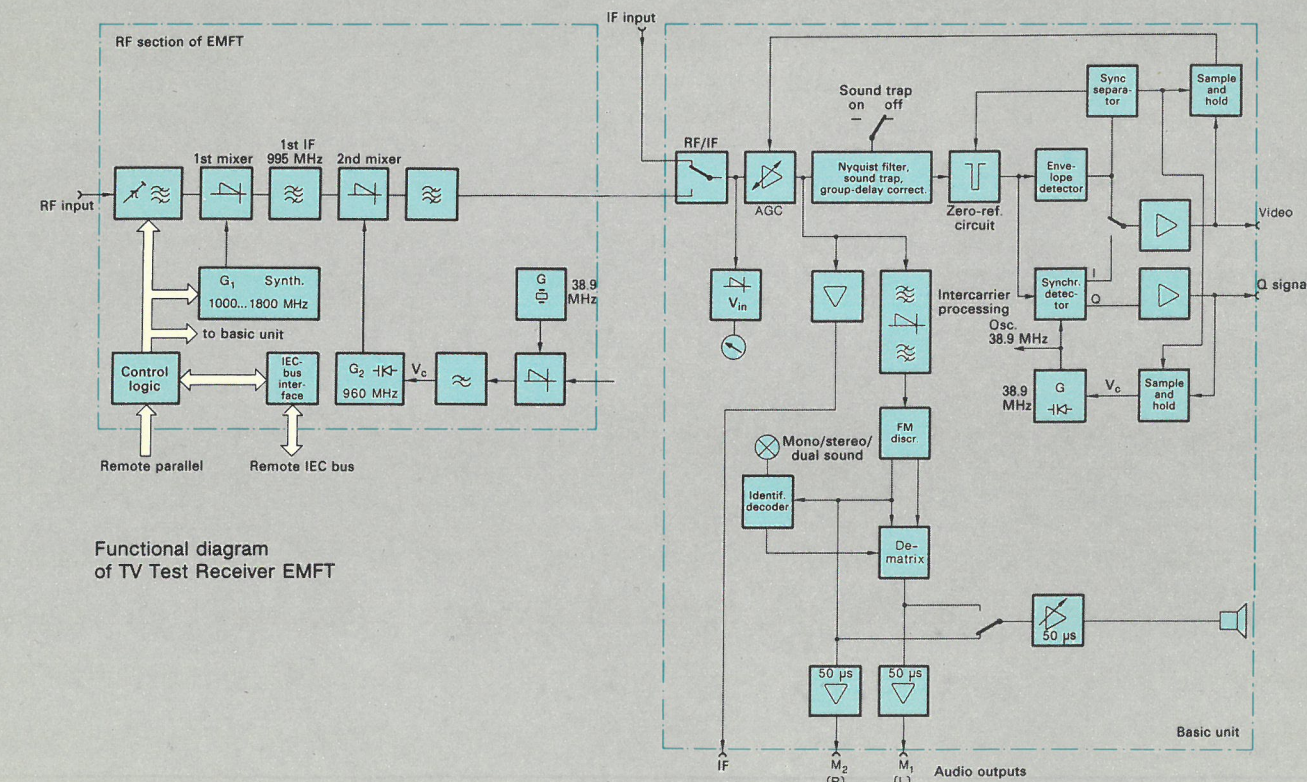


Tolerance of overall amplitude characteristic (RF, IF, VF) without sound trap



Group-delay characteristic of EMF family: curve with sound trap, dashed line for operation without sound trap
EMFT only: Additional ripple with SAW filter $\leq \pm 30$ ns

Tolerance of overall amplitude characteristic (RF, IF, VF) with sound trap
EMFT only: Additional ripple with SAW filter $\leq \pm 0.3$ dB



Functional diagram of TV Test Receiver EMFT

Specifications

	TV Test Receiver EMFT	TV Test Demodulator EMFD	TV Channel Receiver EMFK
Frequency ranges	bands I, II, III, IV/V, special channels up to 470 MHz and IF 38.9 MHz	bands I, II, III, IV/V and IF 38.9 MHz	bands I, II, III, IV/V and IF 38.9 MHz
Channel selection	entry of channel number, automatic search or via IEC bus (option)	fixed-channel operation and tunable with AFC	fixed-channel operation, also AFC when used as demodulator
Frequency processing	by synthesizer	crystal and AFC $\leq \pm 10$ kHz (crystal) $\leq \pm 30$ kHz (AFC)	crystal and AFC $\leq \pm 10$ kHz (crystal) $\leq \pm 30$ kHz (AFC)
Frequency error	$\leq \pm 2.5$ kHz		
Inputs			
RF inputs	BNC female (50 and 75 Ω)	BNC female (50 Ω)	N (50 Ω , RX) and BNC female (50 Ω , demod.)
IF input	BNC female	BNC female	BNC female
Input voltage range ¹⁾			
RF	switchable: 0.15 to 30 mV, 0.5 to 100 mV	20 to 400 mV, 80 to 1.6 mV	250 μ V to 5 mV, 2.5 to 50 mV
IF	5 to 100 mV	5 to 100 mV	5 to 100 mV
Return loss			
IF	≥ 20 dB	≥ 20 dB	≥ 20 dB
RF, 50 Ω	≥ 300 MHz: ≥ 12 dB, > 300 MHz: ≥ 10 dB	≥ 20 dB	≥ 16 dB
RF, 75 Ω	≥ 8 dB		
RF input attenuation	0/10/20 dB automatic or manual adjustment	17/29 dB	0/20 dB (internal link)
Noise figure	VHF: ≤ 9 dB, UHF: ≤ 12 dB (with RF input attenuation 0 dB)	—	typ. 8 dB
Video S/N ratio, rms measurement, CCIR-weighted, HP 10 kHz, ref. to black-to-white transition	≥ 58 dB (input level 3 mV), ≥ 62 dB (input level 10 mV and reduced IF gain)	VHF: ≥ 67 dB, UHF: ≥ 64 dB (input level 400 mV/1.5 V)	≥ 62 dB (input level 5 mV)

Common specifications of equipment family EMF

Transmission characteristics in video channel

Amplitude/frequency response (RF + IF + video, SAW filter switched off in case of EMFT)	see page 6
Group-delay characteristic (SAW filter switched off in case of EMFT)	see page 6
Video S/N ratio	see above
Nonlinearity with modulation depth 10 to 75% (synchronous detection)	
Differential gain	$\leq 3\%$
Differential phase	$\leq \pm 2^\circ$
Tilt (50 Hz)	$\leq 0.5\%$
Gain control, automatic or manual	
Dynamic range	≥ 34 dB (typ. 40 dB) additionally 2×10 dB input attenuation for EMFT

Transmission characteristics in audio channel

Inter-carrier frequency	5.5/5.742 MHz
Frequency response flatness, referred to demphasis 50 μ s (cannot be disabled)	$\leq \pm 0.5$ dB
Harmonic distortion at ± 50 kHz deviation and $f_{mod} = 5$ kHz	$\leq 1\%$, typ. 0.5%
Stereo crosstalk (L $\rightarrow$ R or R $\rightarrow$ L)	down ≥ 36 dB
Channel crosstalk, selective measurement ($M_1 \rightarrow M_2/M_2 \rightarrow M_1$)	down ≥ 70 dB
Inter-carrier S/N ratio, measured to DIN 45405 (quasi-peak) with sinusoidal vision modulation (0 to 5 MHz), referred to nominal output level, weighted to CCIR 468-3	≥ 46 dB (typ. 50 dB)
with all-black picture	≥ 54 dB

SPECIFICATIONS

Outputs

Video outputs	
In-phase signal	2; 75 Ω , BNC (front/rear panel)
Quadrature signal	2; 75 Ω , BNC (front/rear panel)
Output level	
In-phase video output	1 V_{pp} , CVS with standard modulation
Quadrature output	corresponding to 1 V_{pp} with standard modulation and internal phase shift of switching carrier by 90°
IF output	1; 50 Ω , BNC (rear panel)
Output level (AGC)	200 mV _{rms} ± 3 dB
Frequency response flatness in range 33 to 40 MHz, referred to 38.9 MHz	≤ 0.5 dB
Audio outputs	
Signal depending on pilot coding mono or R and L or M ₁ and M ₂	
Front panel	2; BNC, unbalanced, $Z_{out} < 25 \Omega$
Rear panel	37-contact connector
Output level with ± 30 kHz deviation and $f_{mod} = 500$ Hz	+6 dBm ± 0.5 dB

Indication

Analog meter for	a) input level with marking of optimal input level range b) deviation of sound 1 and sound 2 (fsd 50 kHz) c) marker for correct video output level
LED indication for	selected channel, "no vision carrier", "no sound carrier 1", "no sound carrier 2", "no pilot", "stereo", "dual sound"

Miscellaneous

Zero-reference pulse	for checking of residual carrier, field-repetitive, can be enabled in lines 15 and 328 (factory-set) of field blanking interval or triggered by external pulse
Error referred to CVS in case of synchronous detection	$\leq 1.5\%$
Loudspeaker with volume control	can be connected to both sound channels

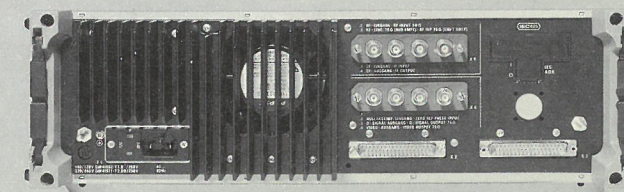
General data

Rated temperature range	+5 to +45 $^\circ$ C
Operating temperature range	0 to +45 $^\circ$ C
Storage temperature range	-40 to +70 $^\circ$ C
Power supply	110/120/220/240 V $\pm 10\%$, 47 to 63 Hz (110 VA)
Dimensions (W $\times$ H $\times$ D), weight	450 mm $\times$ 147 mm $\times$ 525 mm, 16 kg

Ordering information

Order designation	► TV Test Receiver EMFT 821.4019.50 ► TV Test Demodulator EMFD 821.4025.50 ► TV Channel Receiver EMFK 821.4283.50 Bands IV/V 821.4283.53 Bands I/II 821.4283.54 Bands III/III 821.4283.54 Bands II/II please enquire
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¹⁾ Additionally 6 dB beyond lower and upper limits of control range



Rear panel of equipment family EMF



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488 (F so)